

Project Executable Files

DATE	02 JULY 2026
TEAM ID	
PROJECT NAME	A Comprehensive Measure of Well-Being: Human Development Index (HDI) Predictor
MAXIMUM MARKS	3 MARKS

1. Overview

This document guides users through setting up, configuring, and running the HDI Prediction System locally. It details requirements, package dependencies, and troubleshooting tips.

2. Prerequisites

Before running the application, ensure the following are installed:

- **Python:** Version 3.10 or higher.
- **pip:** Python package manager.
- **Git:** For cloning the codebase.

3. Installation and Setup

Step 1: Clone the Repository

Clone the repository and navigate to the project directory:

```
git clone https://github.com/Jaswanth258/HDI.git
cd HDI
```

Step 2: Navigate to the Code Directory

The application source code lives inside the `5.Project Development Phase` folder:

```
cd "5.Project Development Phase/code"
```

Step 3: Create a Virtual Environment

Create a virtual environment named `venv` to keep dependencies isolated:

```
python -m venv venv
```

Step 4: Activate the Virtual Environment

- **Windows (PowerShell):**

```
```powershell  

.\env\Scripts\Activate

```
```

- **Windows (Command Prompt):**

```
```cmd  

.\env\Scripts\activate.bat

```
```

- **macOS / Linux:**

```
```bash  

source env/bin/activate

```
```

Note: If you encounter an execution policy error on Windows PowerShell, resolve it by running:

```
```powershell  

Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned

```
```

Step 5: Install Required Packages

Install dependencies from `requirements.txt`:

```
pip install -r requirements.txt
```

4. Training and Model Execution

Step 1: Train the Machine Learning Model

Train the Linear Regression model and serialize the model and encoder:

```
python ml_model.py
```

Output:

- Preprocesses the dataset `data/hdi\_data.csv`.
- Saves pickled model artifacts to `models/hdi\_model.pkl` and `models/encoder.pkl`.
- Displays model metrics (e.g. RMSE, R^2).

Step 2: Start the Web Application

Start the local Flask development server:

```
python app.py
```

Output:

- Loads the model and encoder.
- Starts the web application on: **`http://127.0.0.1:5000`**

Step 3: Open in Browser

Open your browser and navigate to **`http://127.0.0.1:5000`** to access the dashboard.